

models\cwe_model.py

```
# Type hints for better code documentation and IDE support
from typing import Dict, List, Any, Optional
```

```
class CWEModel:
    """Data model for CWE weakness records"""
```

```
def __init__(self, data: Dict[str, Any]):
    # Extract CWE code (e.g., "CWE-79") with empty string fallback
    self.code = data.get('code', '')
    # Get weakness name/title with empty string fallback
    self.name = data.get('name', '')
    # Extract description text with empty string fallback
    self.description = data.get('description', '')
    # Get list of mitigation strategies with empty list fallback
    self.mitigations = data.get('mitigations', [])
    # Extract examples list with empty list fallback
    self.examples = data.get('examples', [])
    # Get relationships to other CWEs with empty list fallback
    self.relationships = data.get('relationships', [])
    # Track data source with 'unknown' fallback for provenance
    self.source = data.get('source', 'unknown')
```

```
def to_dict(self) -> Dict[str, Any]:
    """Convert to dictionary format"""
    # Return all instance attributes as dictionary for serialization
    return {
        'code': self.code,
        'name': self.name,
        'description': self.description,
        'mitigations': self.mitigations,
        'examples': self.examples,
        'relationships': self.relationships,
        'source': self.source
    }
```

```
def get_id(self) -> str:
    """Get CWE ID number from code"""
    # Check if code follows standard "CWE-123" format
    if self.code.startswith('CWE-'):
        # Return numeric part after "CWE-" prefix (index 4 onwards)
        return self.code[4:]
    # Return original code if format doesn't match
    return self.code
```

```
def get_mitre_url(self) -> str:
    """Get MITRE URL for this CWE"""
    # Extract numeric CWE ID
    cwe_id = self.get_id()
    # Build complete URL to MITRE CWE definition page
    return f"https://cwe.mitre.org/data/definitions/{cwe_id}.html"
```

```
def has_mitigations(self) -> bool:
    """Check if CWE has mitigation strategies"""
    # Verify mitigations exist AND first item isn't placeholder text
    return len(self.mitigations) > 0 and self.mitigations[0] != 'Loading detailed information...'
```